

Predicting FIFA Player Ratings Using Regression Models: A Comprehensive Analysis of Key Attributes and Predictive Accuracy

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Background, Motivation, and Significance

- **Background:** FIFA player ratings are a critical benchmark for assessing player performance and potential. They are used by scouts, managers, and analysts to make key decisions in player development, transfers, and strategy.
- **Significance:**
 - Provides actionable insights into which attributes most influence a player's overall rating.
 - Demonstrates how statistical models can improve understanding and decision-making in sports analytics.
- **Motivation:**
 - Explore how attributes like dribbling, passing, and physical ability impact FIFA ratings.
 - Identify the best predictive model (linear vs. polynomial regression, Ridge, and Lasso) to bridge the gap between data analysis and practical applications.

Research Questions

1. **What attributes most influence FIFA player ratings?**
 - Examining the role of key player skills such as dribbling, passing, and physicality.
2. **How accurately can regression models predict FIFA player ratings?**
 - Testing both linear and non-linear regression approaches for predictive effectiveness.
3. **Which regression model performs best in terms of accuracy and validation?**
 - Comparing performance metrics like R^2 , MSE, and RMSE across linear, polynomial, and regularized models.

Data Set Description

- **Sample Size:** The dataset consists of 16,210 records, representing professional FIFA players.
- **Variables Used:**
 - **Response Variable:** overall (player rating).
 - **Predictor Variables:** wage_eur, age, value_eur, pace, shooting, passing, dribbling, defending, physic, skill_moves, and weak_foot.
- **Handling Missing Values:**
 - Rows with missing values were removed during preprocessing.
 - Total missing values after cleaning: 0.

Pre-Processing

1. **Data Selection:**
 - The analysis focuses on the overall rating as the response variable.

- Key predictors include wage_eur, age, value_eur, pace, shooting, passing, dribbling, defending, physic, skill_moves, and weak_foot.

2. Handling Missing Data:

- Rows with missing values were removed to ensure a complete dataset for analysis.
- After cleaning, no missing values remained, resulting in a dataset ready for analysis.

summary(data)													
Number of missing values after preprocessing: 0													
overall	wage_eur	age	value_eur	pace	shooting	passing	dribbling	defending	physic	skill_moves	weak_foot		
Min. :47.00	Min. : 500	Min. :16.00	Min. : 30000	Min. :27.00	Min. :19.00	Min. :25.00	Min. :28.00	Min. :15.00	Min. :32.00	Min. :2.00	Min. :1.000		
1st Qu.:62.00	1st Qu.: 1000	1st Qu.:21.00	1st Qu.: 525000	1st Qu.:62.00	1st Qu.:42.00	1st Qu.:51.00	1st Qu.:57.00	1st Qu.:38.00	1st Qu.:58.00	1st Qu.:2.00	1st Qu.:3.000		
Median :66.00	Median : 3000	Median :25.00	Median : 1100000	Median :69.00	Median :55.00	Median :58.00	Median :64.00	Median :57.00	Median :66.00	Median :2.00	Median :3.000		
Mean :65.99	Mean : 9047	Mean :25.14	Mean : 2962576	Mean :68.38	Mean :52.56	Mean :57.47	Mean :62.62	Mean :52.08	Mean :64.89	Mean :2.56	Mean :2.995		
3rd Qu.:70.00	3rd Qu.: 8000	3rd Qu.:28.00	3rd Qu.: 2100000	3rd Qu.:76.00	3rd Qu.:63.00	3rd Qu.:64.00	3rd Qu.:69.00	3rd Qu.:64.00	3rd Qu.:72.00	3rd Qu.:3.00	3rd Qu.:3.000		
Max. :91.00	Max. :350000	Max. :39.00	Max. :185000000	Max. :97.00	Max. :93.00	Max. :94.00	Max. :94.00	Max. :89.00	Max. :89.00	Max. :5.00	Max. :5.000		

3. Multicollinearity Assessment:

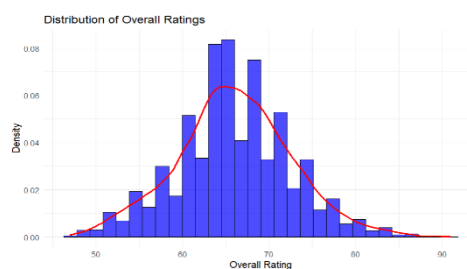
- Variance Inflation Factors (VIF) were computed to evaluate multicollinearity among predictors.
- All VIF values were below the threshold of 10, indicating no significant multicollinearity issues. Thus, all predictors were retained in the model.

vif_values						
wage_eur	age	value_eur	pace	shooting	passing	dribbling
3.581349	1.577773	3.469527	1.690375	5.088973	6.811247	7.326909
defending	physic	skill_moves	weak_foot			
3.738040	1.942615	2.101682	1.136494			

Exploratory Data Analysis (EDA)

1. Distribution Analysis:

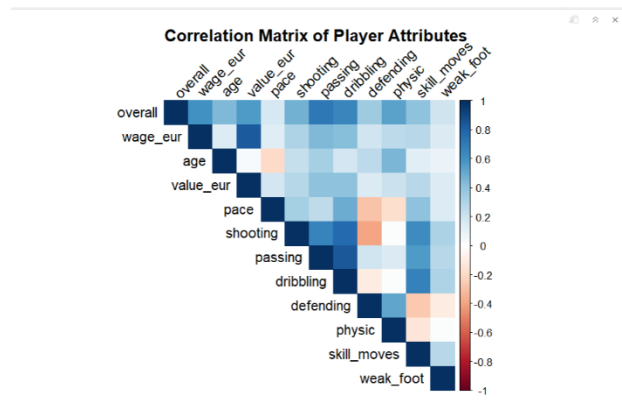
- The histogram of overall ratings shows a roughly normal distribution centered around 65–70.



2. Correlation Analysis:

- A correlation matrix was generated to identify relationships between predictors and the target variable (overall).

- Attributes such as passing, dribbling, and shooting show the strongest correlations with the target variable (overall), while weak_foot and skill_moves have limited impact. This indicates the importance of technical skills in determining player ratings



Methodology

1. Linear Regression:

- Purpose:** Acts as a baseline model to understand the relationships between predictors and the response variable (overall rating).
- Process:**
 - A multiple linear regression model was fitted using all predictors (wage_eur, age, value_eur, pace, shooting, passing, dribbling, defending, physic, skill_moves, weak_foot).

- Results:**

- Adjusted R²: 0.8018
- MSE: 8.9158
- RMSE: 2.9859

Residual standard error: 2.987 on 16205 degrees of freedom
Multiple R-squared: 0.8018, Adjusted R-squared: 0.8016
F-statistic: 5958 on 11 and 16205 DF, p-value: < 2.2e-16

MSE (Linear Regression): 8.91579
RMSE (Linear Regression): 2.985932

- Conclusion:**

- Linear regression performed reasonably but failed to capture complex relationships due to its simplicity.

2. Polynomial Regression:

- Purpose:** To capture non-linear relationships between predictors and the response variable.
- Process:**
 - Added squared terms for key predictors to the dataset to account for quadratic effects.
 - The polynomial regression model was fitted with these additional terms, and the performance metrics were calculated.
- Results:**

- Adjusted R^2 : 0.8883
- MSE: 5.0176
- RMSE: 2.2401
- **Conclusion:**
 - Polynomial regression showed significant improvement over Linear Regression, suggesting the presence of non-linear relationships in the data.

Residual standard error: 2.241 on 16199 degrees of freedom
Multiple R-squared: 0.8884, Adjusted R-squared: 0.8883
F-statistic: 7588 on 17 and 16199 DF, p-value: $< 2.2e-16$

MSE (Polynomial Regression): 5.017598
RMSE (Polynomial Regression): 2.24

3. Ridge Regression:

- **Purpose:** To address multicollinearity and prevent overfitting by applying regularization.
 - **Process:**
 - Applied Ridge regression with cross-validation to determine the optimal regularization parameter (λ).
 - **Results:**
 - Best λ : 0.4857
 - MSE: 7.9558
 - RMSE: 2.8206
- Ridge Regression:
Best Lambda: 0.4857359
MSE: 7.955757
RMSE: 2.820595
- **Conclusion:**
 - While Ridge regression improved multicollinearity but did not outperform Polynomial Regression.

4. Lasso Regression:

- **Purpose:** To perform feature selection by shrinking insignificant predictors to zero.
- **Process:**
 - Lasso regression was applied with cross-validation to determine the optimal λ .
 - Predictors were standardized before fitting the model.
- **Results:**
 - Best λ : 0.0004
 - MSE: 5.0230
 - RMSE: 2.2412

- Some predictors (e.g., weak_foot) were eliminated due to their negligible contributions.

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Lasso Regression:
Best Lambda: 0.0004857359
MSE: 5.022982
RMSE: 2.241201
```

- **Conclusion:**
 - Lasso regression provided comparable performance to Polynomial Regression and improved model interpretability but had slightly higher MSE.

Based on Adjusted R^2 , MSE, and RMSE, Polynomial Regression is identified as the best model.

5. Cross-Validation for the Best Model:

- **Purpose:** To validate the robustness and generalizability of Polynomial Regression.
- **Process:**
 - Performed 10-fold cross-validation on Polynomial Regression (best fit).
- **Results:**
 - Mean MSE: 5.0359
 - Mean RMSE: 2.2437

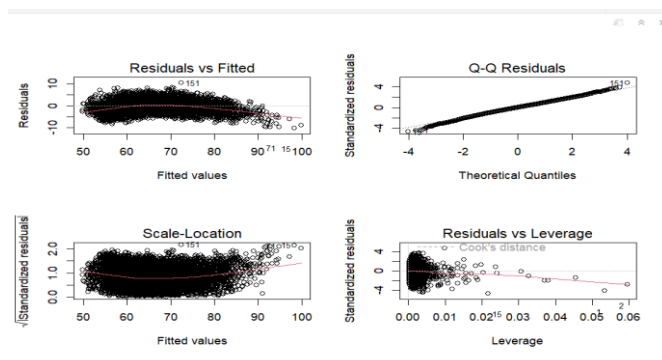
```
Cross-Validation Results for Polynomial Regression:
Mean MSE: 5.035853
Mean RMSE: 2.243714
```

- **Conclusion:**
 - Cross-validation confirms that Polynomial Regression generalizes well to unseen data.

6. Residual Analysis:

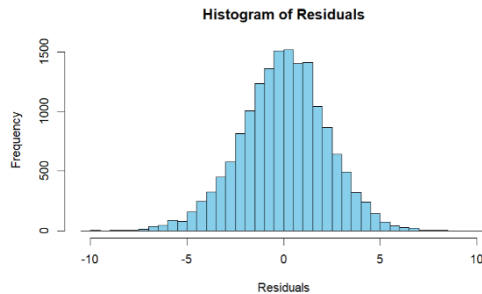
Residual Diagnostics

- The Residuals vs Fitted plot shows residuals are randomly scattered around the horizontal axis, suggesting that the model appropriately captures the relationships in the data.
- The Q-Q plot demonstrates that residuals are mostly aligned with theoretical quantiles, indicating approximate normality with minor deviations at the extremes.
- The Scale-Location plot shows a relatively consistent spread, confirming homoscedasticity.



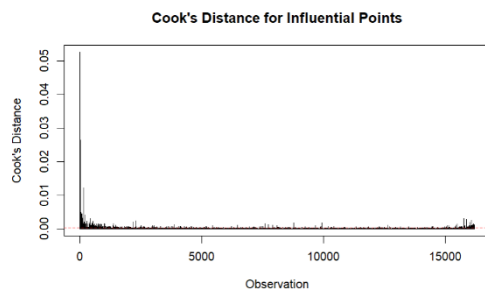
- The Residuals vs Leverage plot identifies potential influential points. Observations beyond Cook's distance threshold were further examined and removed to refine the model.

Histogram of Residuals



- The histogram of residuals depicts a roughly normal distribution centered around zero, validating the assumption of normally distributed residuals.

Cook's Distance for Influential Points



- The Cook's distance plot highlights observations with high leverage that could influence the model. Points exceeding the threshold were removed from the dataset. After excluding these high-leverage points, the model's performance improved significantly.

7. Polynomial Regression with Cleaned Data Based on Cook's Distance

Residual standard error: 1.895 on 15125 degrees of freedom
Multiple R-squared: 0.9069, Adjusted R-squared: 0.9068
F-statistic: 8667 on 17 and 15125 DF, p-value: < 2.2e-16

Polynomial Regression (Cleaned Data):
MSE: 3.588398
RMSE: 1.894307
Cross-Validation Results (Cleaned Data):
Mean MSE: 3.634495
Mean RMSE: 1.906424

- After removing high-leverage points, the refined polynomial regression model demonstrated the following improvements:
 - Mean Squared Error (MSE): Reduced to 3.588398 (from 5.017598 before cleaning).
 - Root Mean Squared Error (RMSE): Reduced to 1.894307 (from 2.241201 before cleaning).
 - Cross-Validation Results:
 - Mean MSE: 3.634495.
 - Mean RMSE: 1.906424.

- The adjusted R^2 increased to 0.9068, indicating that the model now explains more variance in the response variable.
- The quadratic terms (e.g., $I(\text{pace}^2)$, $I(\text{shooting}^2)$, $I(\text{passing}^2)$) were confirmed to have significant contributions to the overall model.

Results

Comparison of Model Performance Metrics

Method	Adjusted R^2	MSE	RMSE	Cross-Validated RMSE
Linear Regression	0.8016	8.9158	2.9859	-
Polynomial Regression	0.8883	5.0176	2.2401	2.2437
Ridge Regression	-	7.9558	2.8206	-
Lasso Regression	-	5.0230	2.2412	-
Polynomial (Cleaned)	0.9068	3.5884	1.8943	1.9064

Polynomial Regression (Cleaned Data based on Cook's Distance) emerged as the best-performing Model:

- Highest **Adjusted R^2** (0.9068) and lowest **MSE** (3.5884) and **RMSE** (1.8943).
- Cross-validation confirmed robustness, with a Cross-Validated RMSE of **1.9064**.

Linear Regression served as a baseline model but showed weaker performance due to limited capture of non-linear relationships.

Ridge and Lasso Regression mitigated multicollinearity but underperformed compared to Polynomial Regression.

Conclusion

The Polynomial Regression model, refined by removing high-leverage points, emerged as the most effective approach for predicting FIFA player ratings. With its high Adjusted R^2 (0.9068) and low RMSE (1.8943), this model successfully captured the non-linear relationships between player attributes and their ratings. The inclusion of quadratic terms, such as pace^2 , shooting^2 , and passing^2 , highlights the importance of these predictors in identifying peak potential. This methodology provides valuable insights for talent assessment, enabling teams to evaluate player potential and monitor performance growth.